



Medical Terminology

Anatomical Position, Directional Terms and Movements

Dr. Rawand Polus Shamoon
PhD Hematopathology

What we will cover...

Content Objectives

- Students will be able to gain a better understanding and application of medical terminology in relation to Anatomical Planes, Directions and Movements.



- **Language Objectives**

- Students will be able to define basic anatomical terms and apply those terms when dissecting.
- Students will be able to compare the differences between directional terms and terms associated with movement.

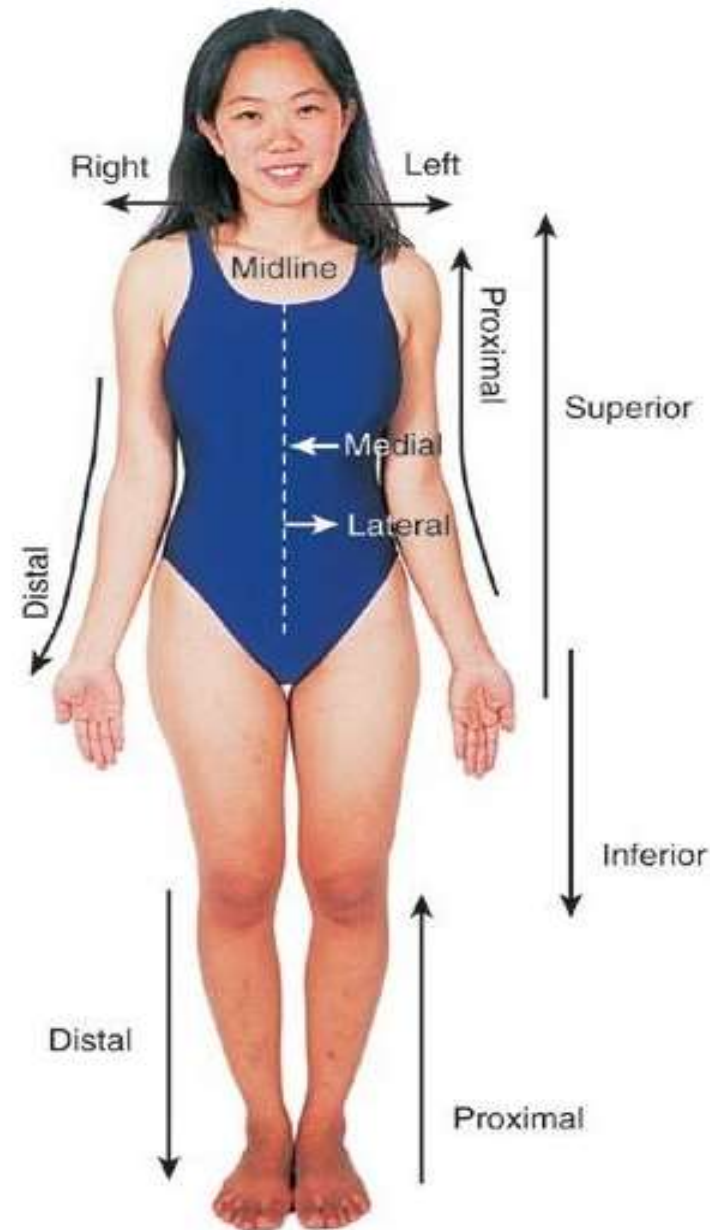
Anatomical Position

- Standing erect, with palms and feet facing forward
- Is the standard reference point in which all positions, movements, and planes are described



Positions and Directions

Terms of position and direction describe the position of one body part relative to another



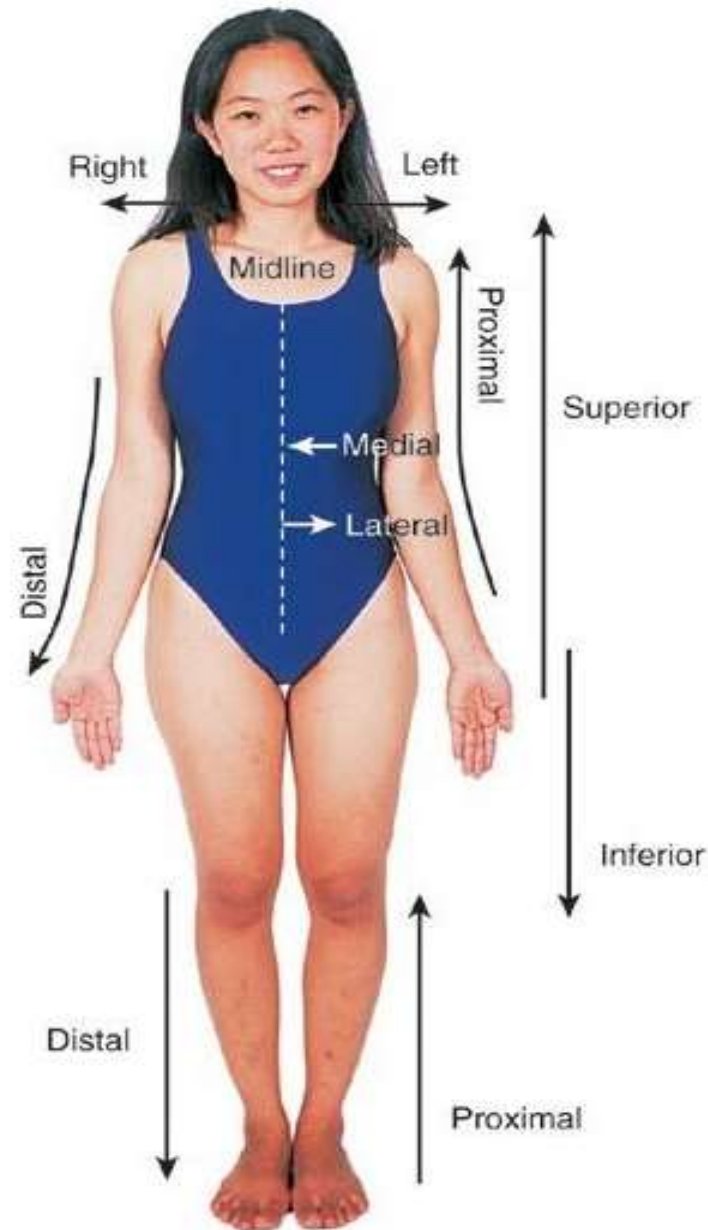
Positions and Directions

Superior

- Refers to a structure being closer to the head or higher than another structure in the body

Inferior

- Refers to a structure being closer to the feet or lower than another structure in the body



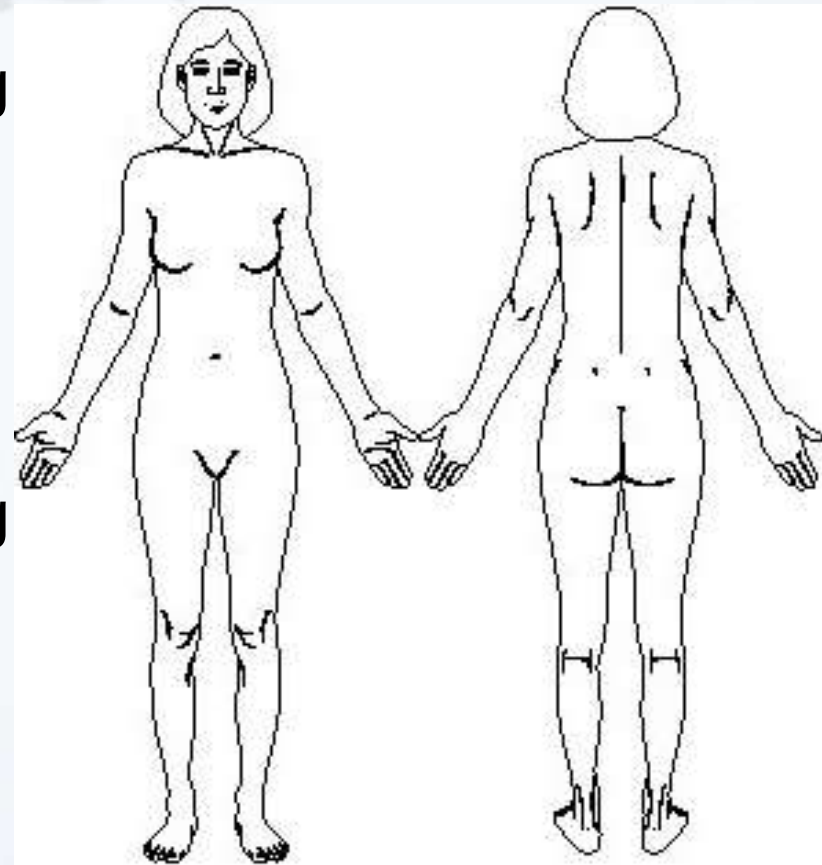
Positions and Directions

Anterior

- Refers to a structure being more in front than another structure in the body

Posterior

- Refers to a structure being more in back than another structure in the body



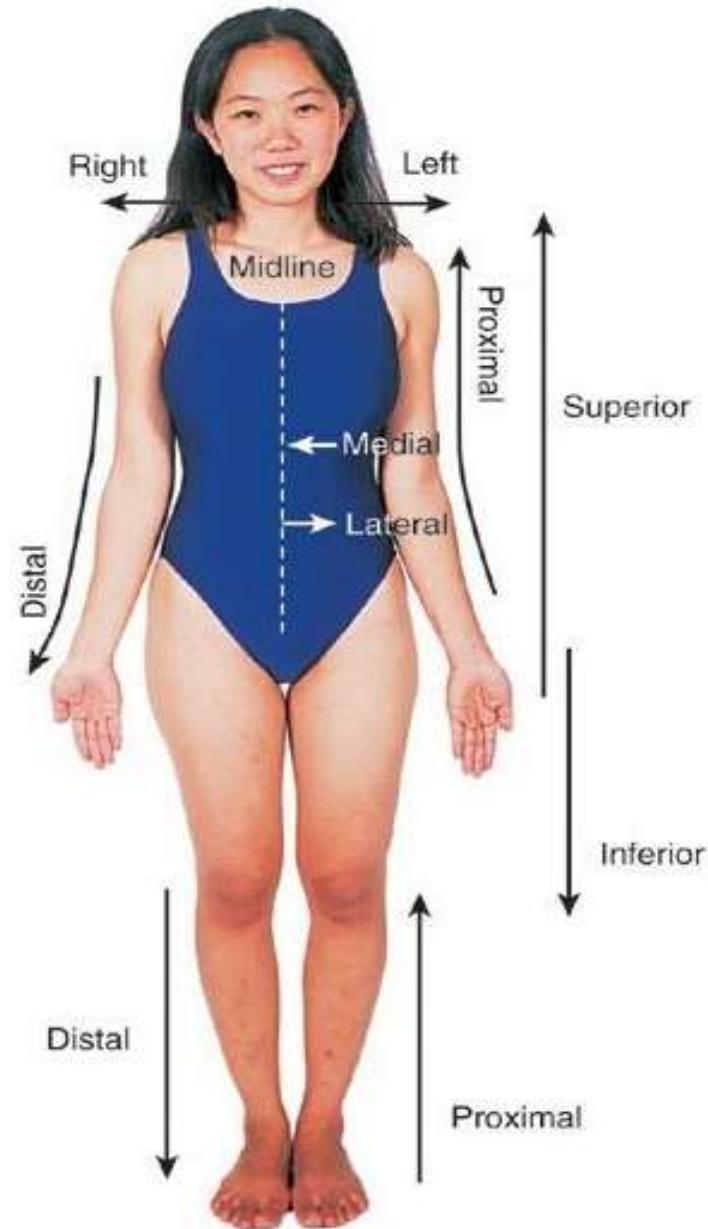
Positions and Directions

Medial

- Refers to a structure being closer to the midline or median plane of the body than another structure of the body

Lateral

- Refers to a structure being farther away from the midline than another structure of the body

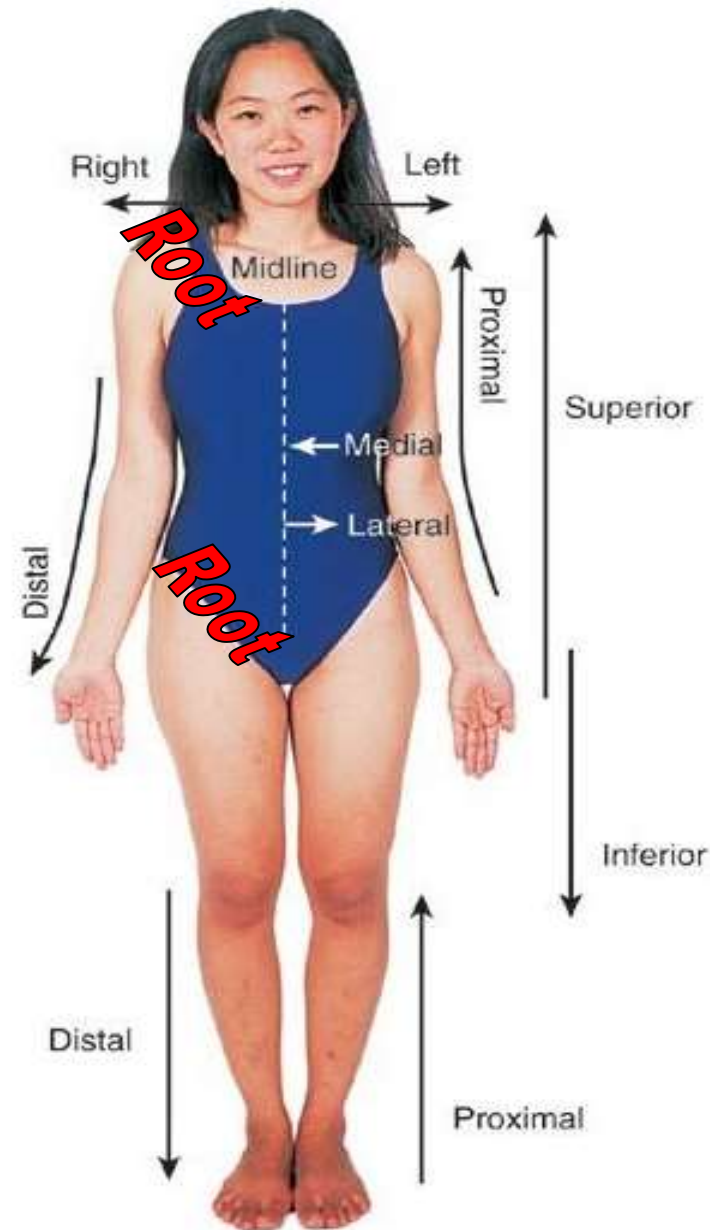


Positions and Directions

Distal

(Reference to the extremities only)

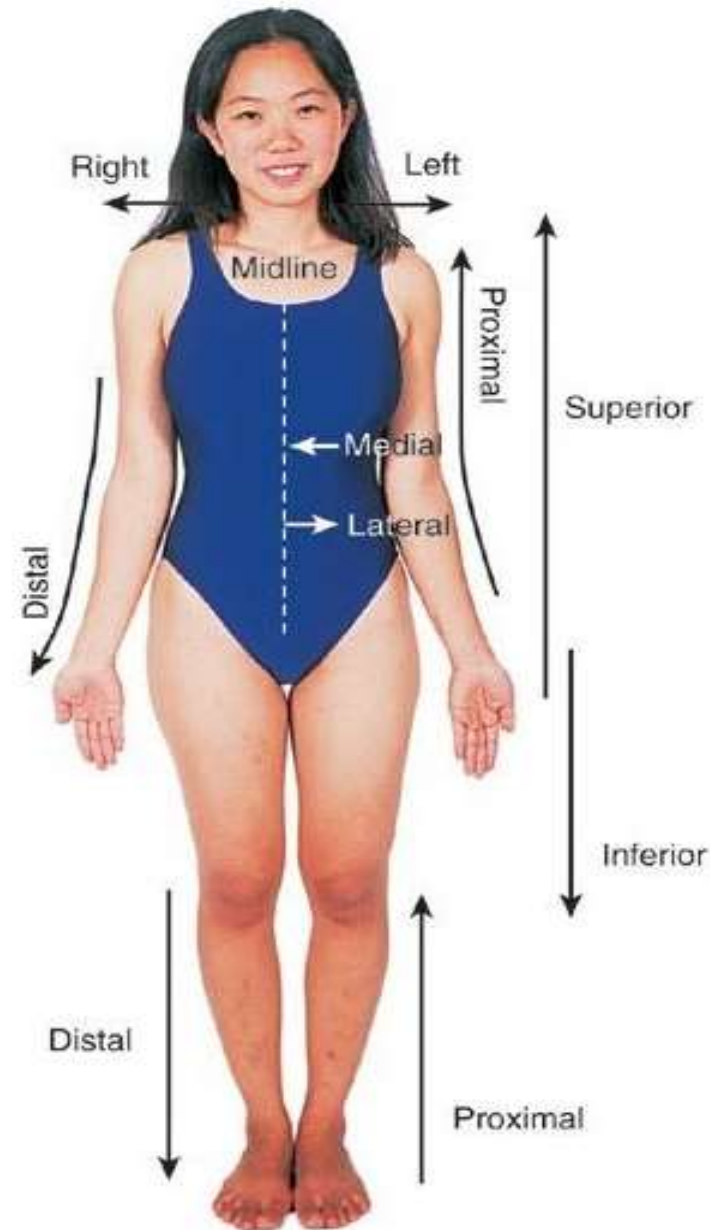
- Refers to a structure being further away from the root of the limb than another structure in the limb



Proximal

(Reference to the extremities only)

- Refers to a structure being closer to the root of the limb than another structure in that limb



Distal / Proximal Cont.

- When you divide the skeleton into **Axial** (Blue) and **Appendicular** (Yellow) you can better understand the extremities and their roots.

Root ————— Proximal
————— Distal



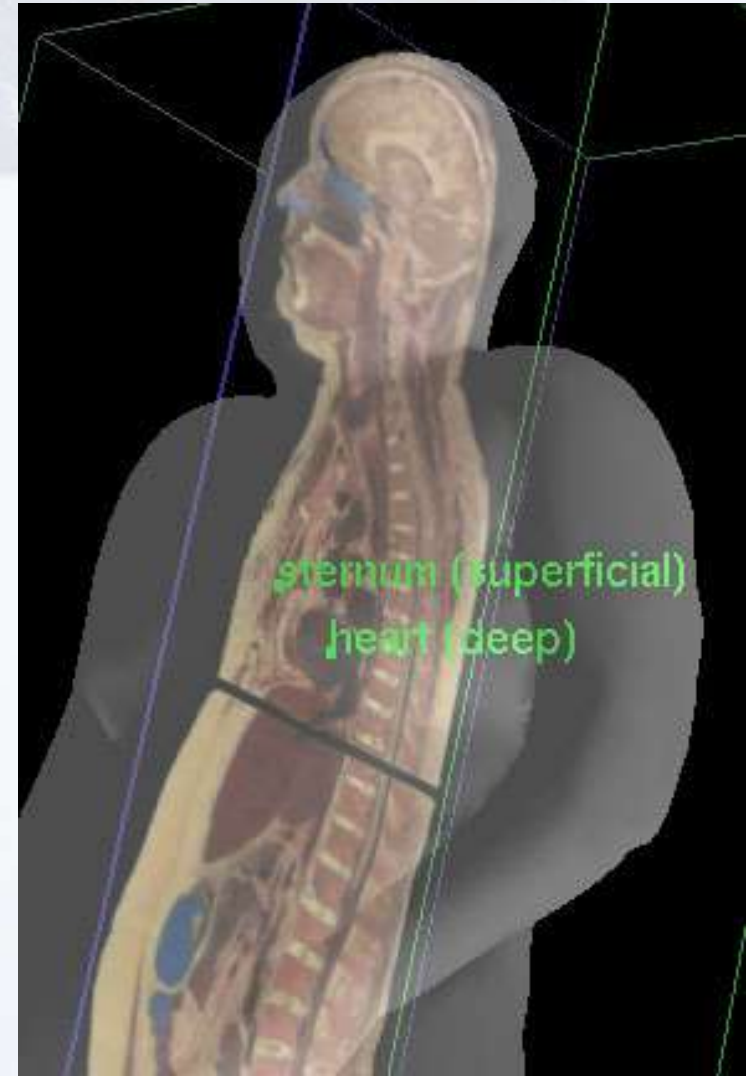
Positions and Directions

Superficial

- Refers to a structure being closer to the surface of the body than another structure

Deep

- Refers to a structure being closer to the core of the body than another structure



Positions and Directions

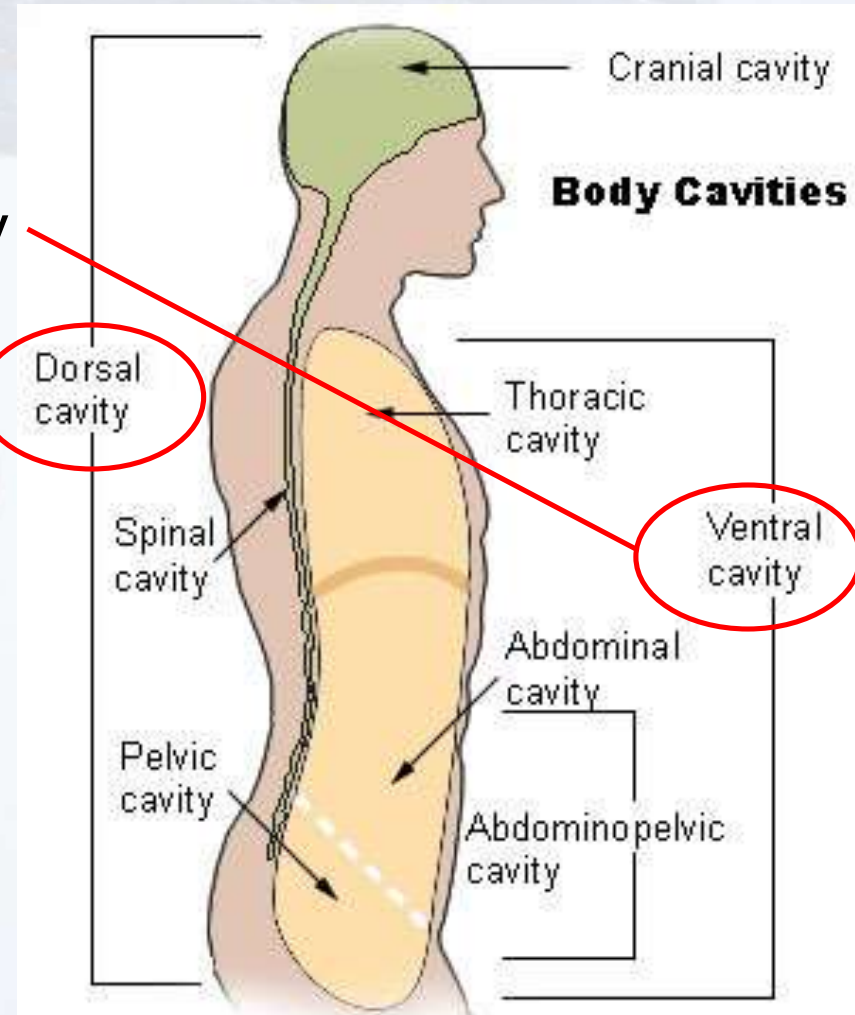
Ventral

- Towards the front or belly
 - You Vent out or your nose and mouth.



Dorsal

- Towards the back
 - Like the Dorsal fin of a dolphin.



Positions and Directions

Prone

- Lying face down
 - Like a Pro Baseball player sliding into Home.



Prone



Supine

- Lying face up
 - Lying on your spine and you can have soup poured into your mouth.



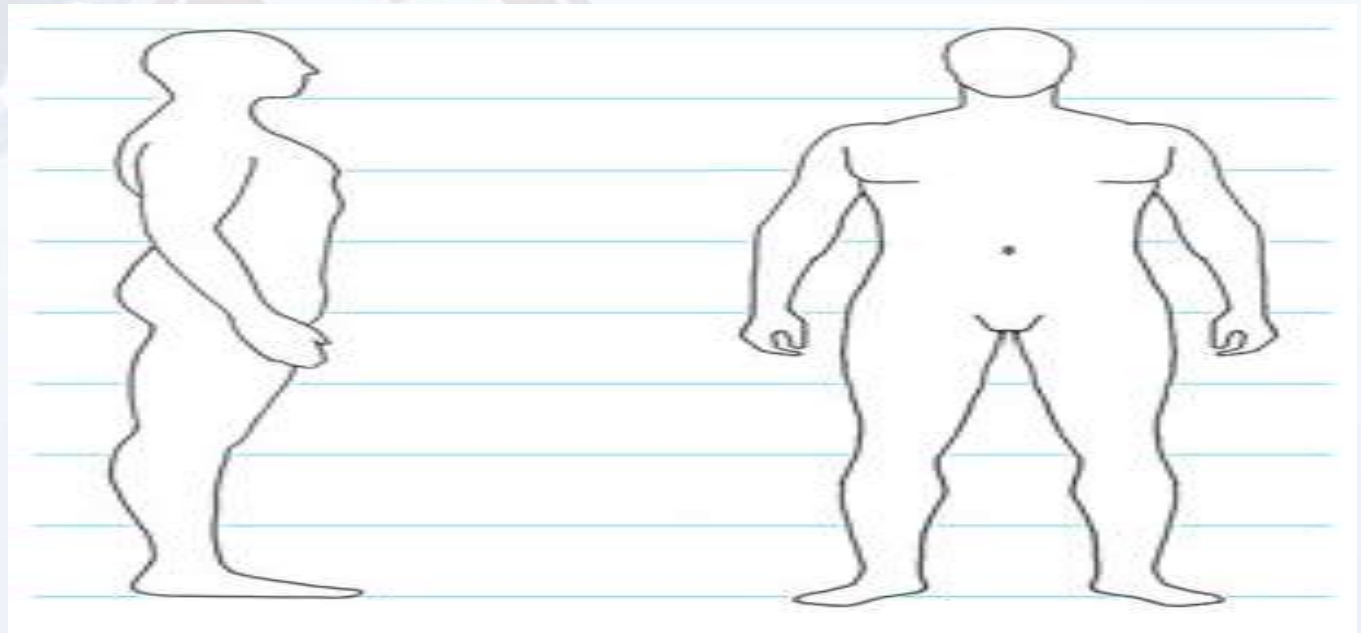
Supine

Unilateral

- Pertaining to one side of the body

Bilateral

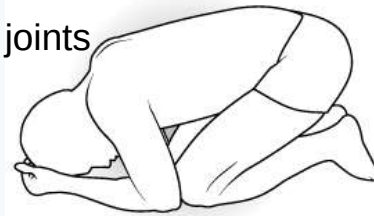
- Pertaining to both sides of the body



Movements

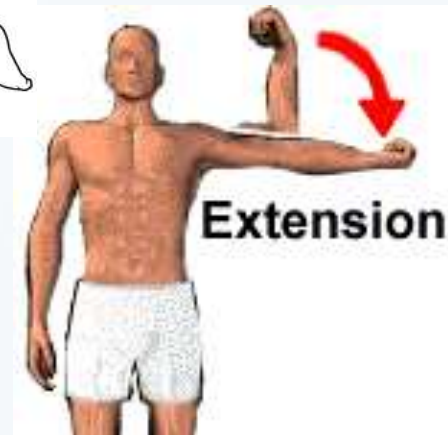
Flexion

- Bending a joint or decreasing the angle between two bones
 - In the Fetal Position we are flexing our joints



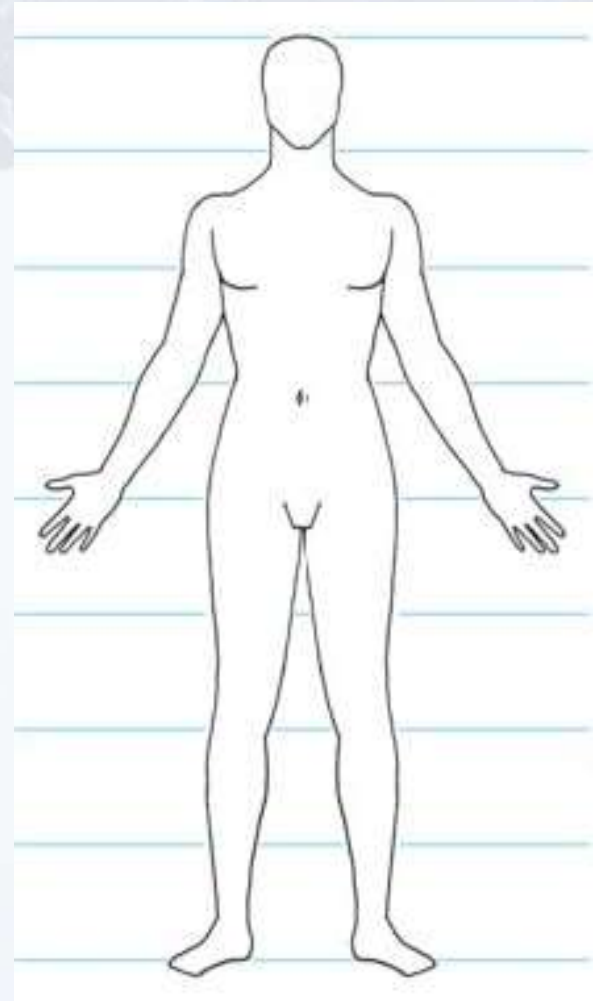
Extension

- Straightening a joint or increasing the angle between two bones
 - In the Anatomical Position we are extending our joints



Hyperextension

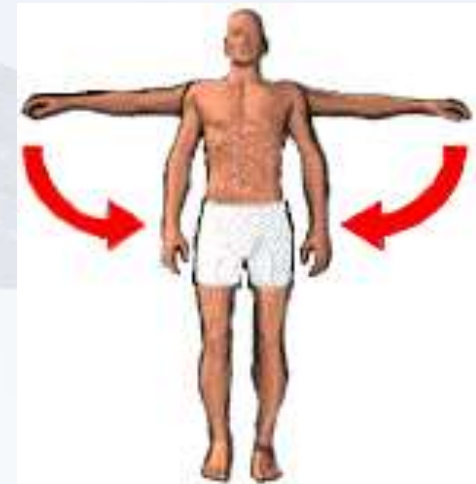
Excessive extension of the parts at a joint beyond anatomical position.



Movements

Adduction

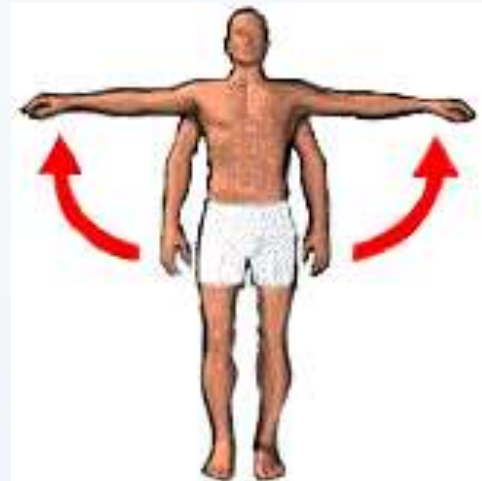
- Moving a body part towards the midline of the body



Adduction

Abduction

- Moving a body part away from the midline of the body

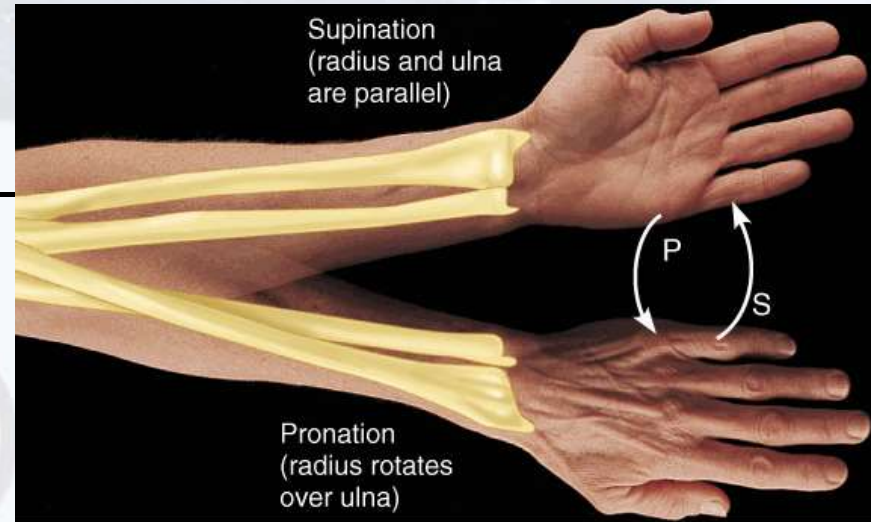


Abduction

Movements

Pronation

- Turning the arm or foot downward
- (palm or sole of the foot down)
 - Prone



Supination

- Turning the arm or foot upward
- (palm or sole of the foot - up)
 - Supine



Movements

Retraction

- Moving a part backward

Protraction

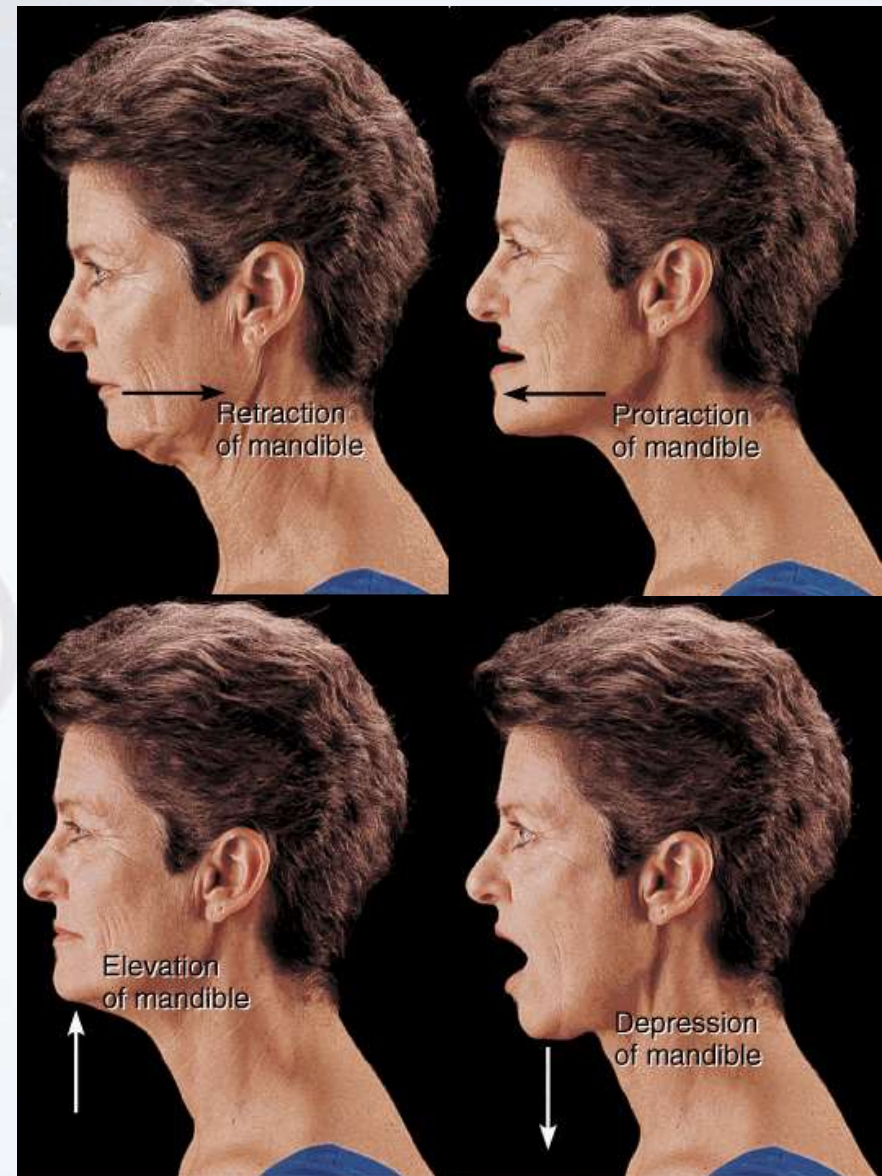
- Moving a part forward

Elevation

- Raising a part

Depression

- Lowering a part



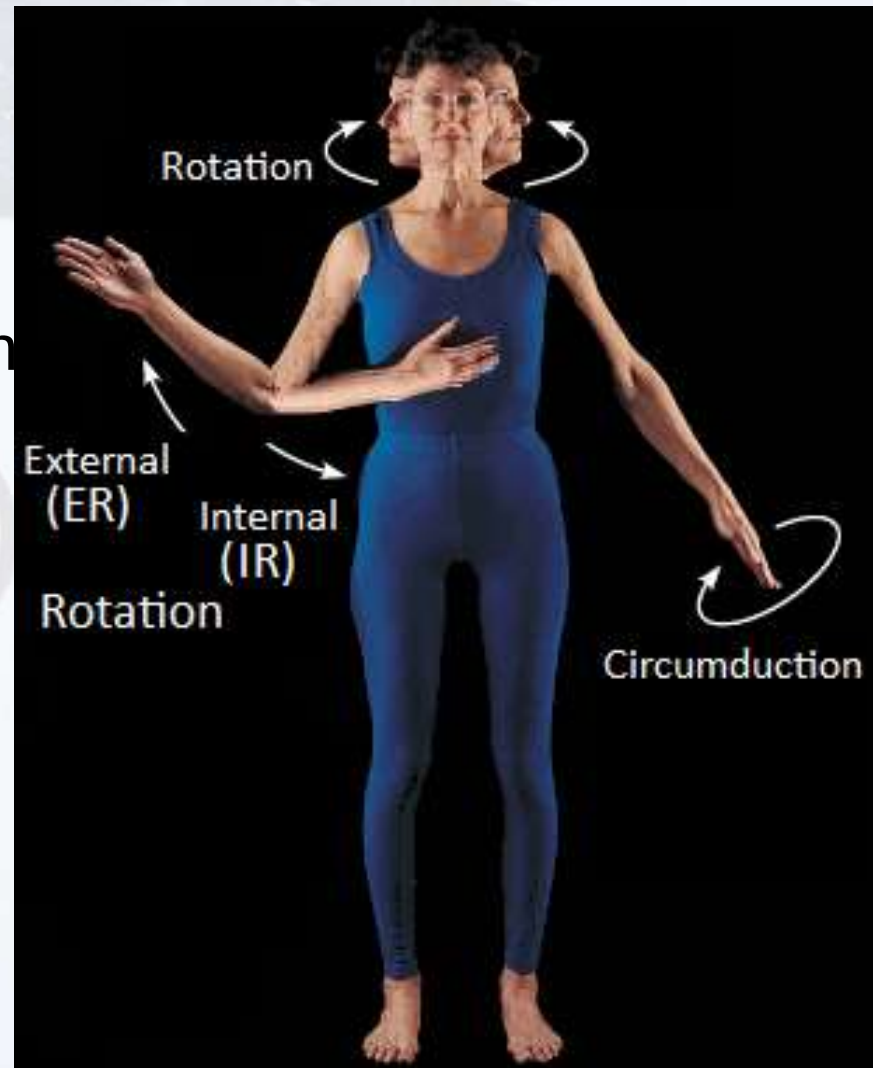
Movements

Rotation

- Turning on a single axis

Circumduction

- Tri-planar, circular motion at the hip or shoulder

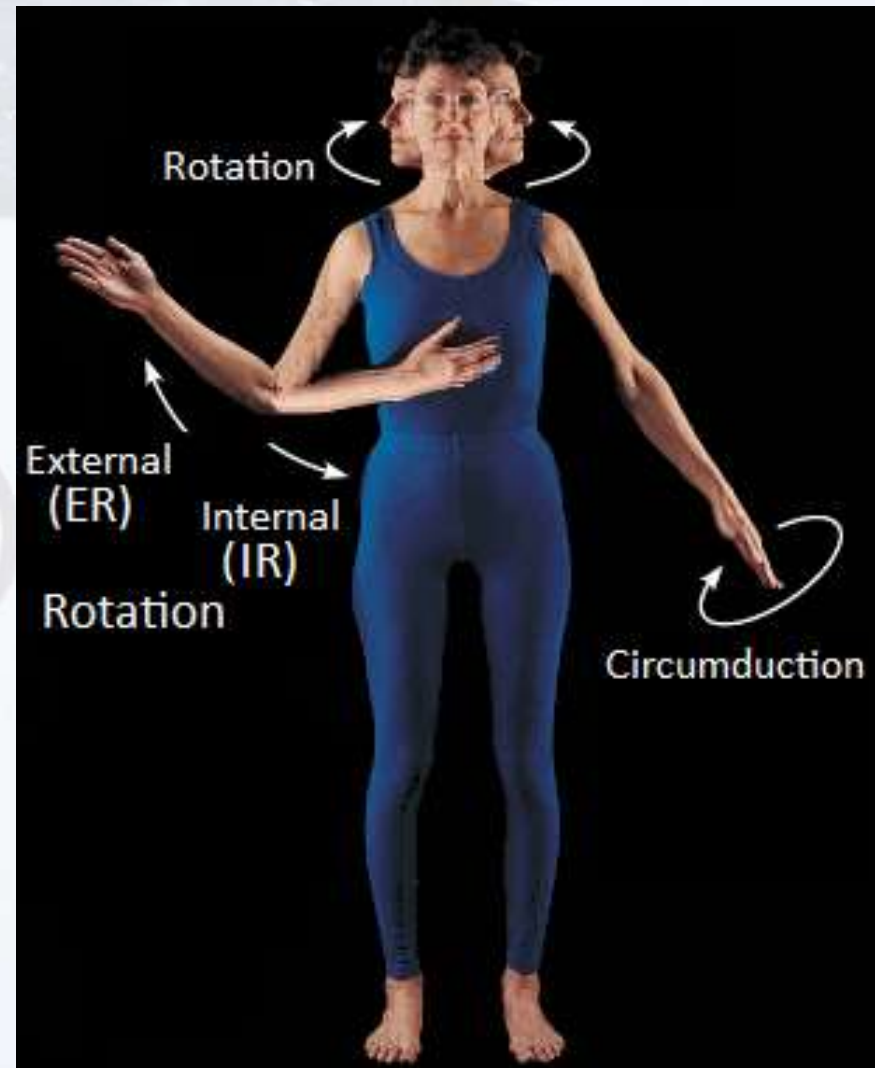


Internal rotation

- Rotation of the hip or shoulder toward the midline

External rotation

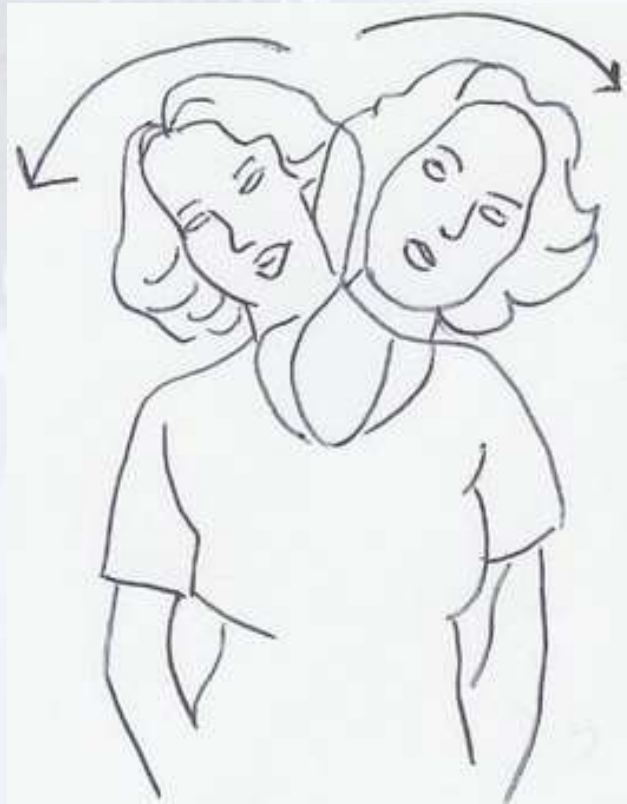
- Rotation of the hip or shoulder away from the midline



Movements

Lateral Flexion

- Side-bending left or right



Movements of the Foot

Inversion

- Turning the sole of the foot inward

Eversion

- Turning the sole of the foot outward

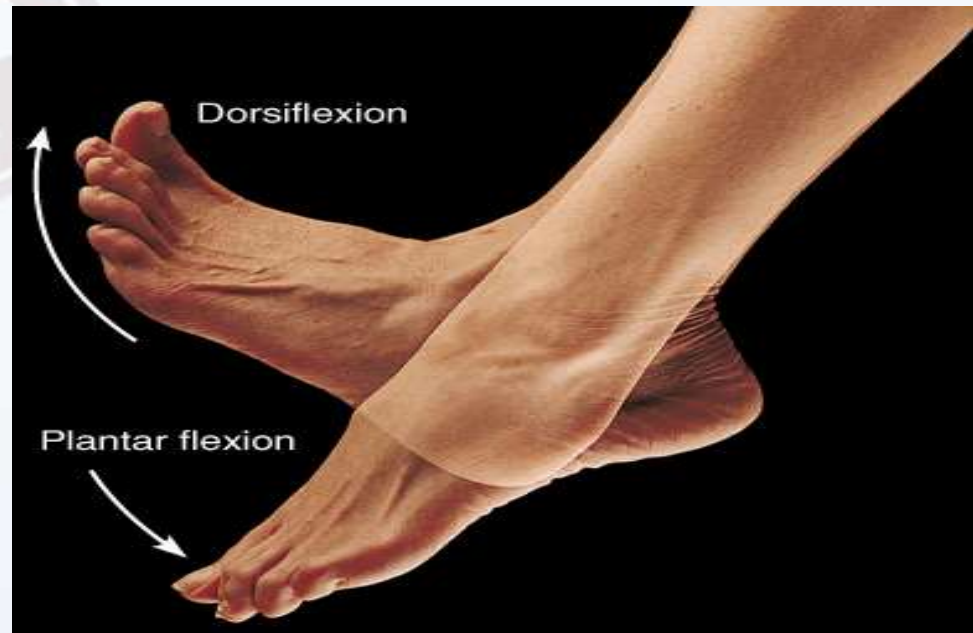


Dorsiflexion

- Ankle movement bringing the foot towards the shin

Plantarflexion

- Ankle movement pointing the foot downward



Movements of the Wrist & Thumb

Radial Deviation

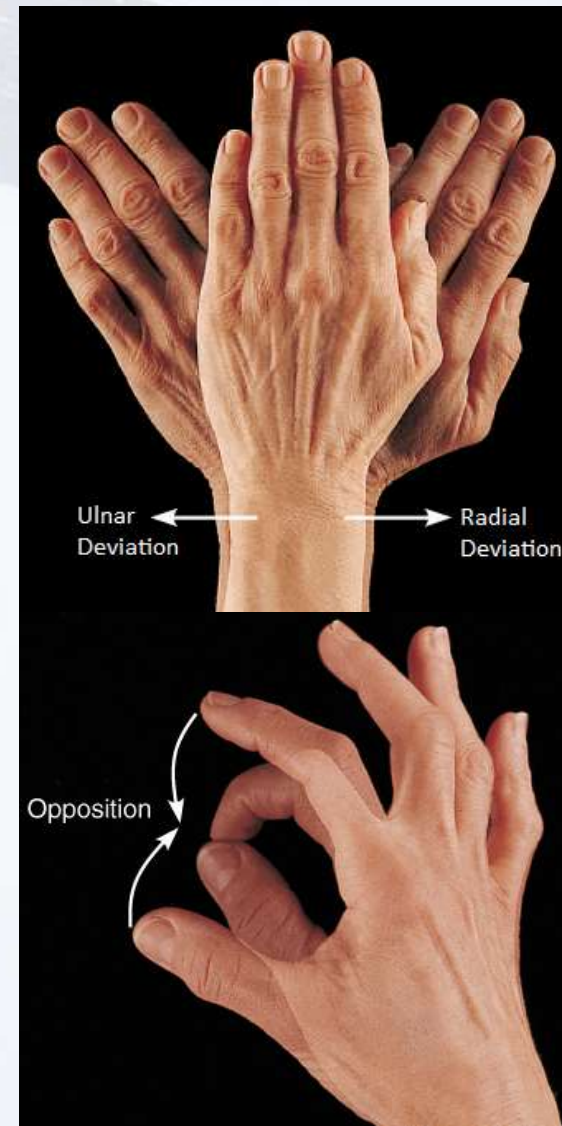
- Movement of the wrist towards the radius or lateral side.

Ulnar Deviation

- Movement of the wrist towards the ulna or medial side.

Opposition

- Movement of the thumb across the palm of the hand.





THANK YOU